Exponentials, Radicals, Polynomials & Rational Expressions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01 • 20 SPRs + 25 MCQs • Difficulty Progression • Official Digital SAT Curriculum

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Student-Produced Response Questions (SPRs): Foundational Fluency (1–7)

1 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

If $f(x) = 5 \cdot 4^x$, what is $f(0)$?

Answer: 5. Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $f(0) = 5 \cdot 4^0 = 5 \cdot 1 = 5$ Therefore, the verified correct option is 5.

2 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

If $3^x = 81$, what is the value of x?

Answer: 4. $81 = 3^4$, so $3^x = 3^4 \Rightarrow x = 4$.

3 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

What is the value of $8^{2/3}$?

Answer: 4. $8^{2/3} = (\sqrt[3]{8})^2 = 2^2 = 4$.

4 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

If $f(x) = 12(1.5)^x$, what is the value of $f(2)$?

Answer: 27. $f(2) = 12(1.5)^2 = 12(2.25) = 27$.

5 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

What is the degree of the polynomial $5x^4 - 2x^3 + 7x - 9$?

Answer: 4. The degree of a polynomial in one variable is the highest exponent, which is 4.

6 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

When $(x + 4)(x + 1)$ is expanded, what is the coefficient of x?

Answer: 5. $(x + 4)(x + 1) = x^2 + 5x + 4$, so the coefficient of x is 5.

7 [Foundational] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Simplify $\frac{x^2 - 2x - 35}{(x + 5)}$, then evaluate the result at $x = 2$.

Answer: -5. $\frac{(x - 7)(x + 5)}{(x + 5)} = (x - 7)$ (for $x \neq -5$). At $x = 2$: -5.

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Part I — Student-Produced Response Questions (SPRs): Intermediate Application (8–14)

8 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

If $4^x = 64$, what is x?

Answer: 3. Express 64 as a power with base 4: $64 = 4^3$. Equating exponents in $4^x = 4^3$ gives $x = 3$.

9 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

When $(x + 1)(x + 6)$ is expanded, what is the coefficient of x?

Answer: 7. $(x + 1)(x + 6) = x^2 + 7x + 6$, so the coefficient of x is 7.

10 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

A quantity $A_0 = 800$ increases by 25% to A_1 and then decreases by 20% to A_2. What is the final quantity $n = A_2$?

Answer: -1. Computing $800 \times (1 + 0.25) \times (1 - 0.20) = 800$.

11 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

When $(x - 1)(x + 8)$ is expanded, what is the constant term?

Answer: -8. $(x - 1)(x + 8) = x^2 + 7x - 8$, so the constant term is -8.

12 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

When $(x + 4)(x + 4)$ is expanded, what is the constant term?

Answer: 16. $(x + 4)(x + 4) = x^2 + 8x + 16$, so the constant term is 16.

13 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

When $(x + 1)(x + 7)$ is expanded, what is the constant term?

Answer: 7. $(x + 1)(x + 7) = x^2 + 8x + 7$, so the constant term is 7.

14 [Intermediate] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Simplify $\frac{x^2 + 10x + 16}{(x + 2)}$, then evaluate the result at $x = 0$.

Answer: 8. $\frac{(x + 8)(x + 2)}{(x + 2)} = (x + 8)$ (for $x \neq -2$). At $x = 0$: 8.

Part I — Student-Produced Response Questions (SPRs): Challenging Synthesis (15–20)

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15 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
If $f(x) = 5 \cdot 2^x$, what is $f(4)$?

Answer: 80. Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $f(4) = 5 \cdot 2^4 = 5 \cdot 16 = 80$ Therefore, the verified correct option is 80.

16 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
What is the value of $16^{3/2}$?

Answer: 64. $16^{3/2} = (\sqrt{16})^3 = 4^3 = 64$.

17 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
When $(x + 3)(x - 2)$ is expanded, what is the coefficient of x?

Answer: 1. $(x + 3)(x - 2) = x^2 + x - 6$, so the coefficient of x is 1.

18 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
Simplify $\frac{x^2 + 5x - 14}{(x + 7)}$, then evaluate the result at $x = 4$.

Answer: 2. $\frac{(x - 2)(x + 7)}{(x + 7)} = (x - 2)$ (for $x \neq -7$). At $x = 4$: 2.

19 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
When $(x - 6)(x + 9)$ is expanded, what is the constant term?

Answer: -54. $(x - 6)(x + 9) = x^2 + 3x - 54$, so the constant term is -54.

20 [Challenging] • [SPR] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
Simplify $\frac{x^2 + 11x + 24}{(x + 3)}$, then evaluate the result at $x = 1$.

Answer: 9. $\frac{(x + 8)(x + 3)}{(x + 3)} = (x + 8)$ (for $x \neq -3$). At $x = 1$: 9.

Part II — Multiple-Choice Questions (MCQs): Foundational Fluency (21–28)

21 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials
$(2x^2 + 3x - 1) - (x^2 - 2x + 4)$ simplifies to

a. $x^2 + 5x - 5$ ✓

b. $3x^2 + x + 3$

c. $x^2 - 5x + 5$

d. $x^2 + x - 5$

Distribute the negative sign: $2x^2 + 3x - 1 - x^2 + 2x - 4 = (2x^2 - x^2) + (3x + 2x) + (-1 - 4) = x^2 + 5x - 5$.

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22 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(-8x^4)(-5x^4)$?

a. $40x^8$ ✓

b. $40x^9$

c. $40x^{16}$

d. $-13x^8$

Multiply coefficients and add exponents: $-8 \cdot -5 = 40$, $x^4 \cdot x^4 = x^8$.

23 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $x^9 \cdot x^2$?

a. x^{18}

b. x^{11} ✓

c. $2x^{11}$

d. x^7

Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $x^9 \cdot x^2 = x^{9+2} = x^{11}$

Therefore, the verified correct option is x^{11}.

24 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Add: $(7x^2 - x + 10) + (4x^2 + 4x - 10)$

a. $13x^2 + 3x$

b. $11x^2 + 3x + 2$

c. $11x^2 + 5x$

d. $11x^2 + 3x$ ✓

Combine like terms to get $11x^2 + 3x$.

25 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(b^8)^2$?

a. b^{16} ✓

b. b^{18}

c. b^{64}

d. b^{10}

Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $(b^8)^2 = b^{8 \cdot 2} = b^{16}$

Therefore, the verified correct option is b^{16}.

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26 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(b^7)^5$?

a. b^{35} ✓

b. b^{13}

c. b^{37}

d. b^{12}

Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $(b^7)^5 = b^{7 \cdot 5} = b^{35}$

Therefore, the verified correct option is b^{35}.

27 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(a^6)^8$?

a. a^{14}

b. a^{50}

c. a^{48} ✓

d. a^{15}

Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $(a^6)^8 = a^{6 \cdot 8} = a^{48}$

Therefore, the verified correct option is a^{48}.

28 [Foundational] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(y^8)^5$?

a. y^{14}

b. y^{42}

c. y^{13}

d. y^{40} ✓

Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $(y^8)^5 = y^{8 \cdot 5} = y^{40}$

Therefore, the verified correct option is y^{40}.

Part II — Multiple-Choice Questions (MCQs): Intermediate Application (29–37)

29 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Subtract: $(x^2 - 2x) - (4x^2 - 8x - 9)$

a. $-3x^2 + 6x + 11$

b. $-3x^2 + 8x + 9$

c. $-x^2 + 6x + 9$

d. $-3x^2 + 6x + 9$ ✓

Combine like terms to get $-3x^2 + 6x + 9$.

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30 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Subtract: $(5x^2 - 4x + 3) - (4x^2 + 4x - 3)$

a. $x^2 - 8x + 6$ ✓

b. $x^2 - 8x + 8$

c. $3x^2 - 8x + 6$

d. $x^2 - 6x + 6$

Combine like terms to get $x^2 - 8x + 6$.

31 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(-6x^1)(8x^3)$?

a. $-48x^5$

b. $2x^4$

c. $-48x^3$

d. $-48x^4$ ✓

Multiply coefficients and add exponents: $-6 \cdot 8 = -48$, $x^1 \cdot x^3 = x^4$.

32 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Add: $(7x^2 - 7x - 2) + (2x^2 + 3x - 6)$

a. $9x^2 - 2x - 8$

b. $9x^2 - 4x - 8$ ✓

c. $9x^2 - 4x - 6$

d. $11x^2 - 4x - 8$

Combine like terms to get $9x^2 - 4x - 8$.

33 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Subtract: $(6x^2 + 5x - 3) - (2x^2 + 2x - 5)$

a. $6x^2 + 3x + 2$

b. $4x^2 + 3x + 4$

c. $4x^2 + 5x + 2$

d. $4x^2 + 3x + 2$ ✓

Combine like terms to get $4x^2 + 3x + 2$.

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34 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(x^6)^9$?

a. x^{16}

b. x^{56}

c. x^{54} ✓

d. x^{15}

Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $(x^6)^9 = x^{6 \cdot 9} = x^{54}$

Therefore, the verified correct option is x^{54}.

35 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $b^7 \cdot b^4$?

a. b^{28}

b. b^3

c. $2b^{11}$

d. b^{11} ✓

Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $b^7 \cdot b^4 = b^{7+4} = b^{11}$

Therefore, the verified correct option is b^{11}.

36 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Subtract: $(5x^2 - 8x - 7) - (6x^2 - 3x + 9)$

a. $-x^2 - 5x - 16$ ✓

b. $-x^2 - 5x - 14$

c. $x^2 - 5x - 16$

d. $-x^2 - 3x - 16$

Combine like terms to get $-x^2 - 5x - 16$.

37 [Intermediate] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $x^{1/3}$ for $x \geq 0$?

a. $3x$

b. $\frac{x}{3}$

c. $\sqrt[3]{x}$ ✓

d. $\frac{1}{x^3}$

By definition of rational exponents, $x^{1/n} = \sqrt[n]{x}$. Thus $x^{1/3} = \sqrt[3]{x}$.

Part II — Multiple-Choice Questions (MCQs): Challenging Synthesis (38–45)

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38 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(-4x^4)(-6x^2)$?

a. $24x^7$

b. $-10x^6$

c. $24x^6$ ✓

d. $24x^8$

Multiply coefficients and add exponents: $-4 \cdot -6 = 24$, $x^4 \cdot x^2 = x^6$.

39 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $x^4 \cdot x^7$?

a. x^3

b. x^{28}

c. x^{11} ✓

d. $2x^{11}$

Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $x^4 \cdot x^7 = x^{4+7} = x^{11}$

Therefore, the verified correct option is x^{11}.

40 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Add: $(8x^2 - 4x + 3) + (2x^2 - 7x - 8)$

a. $10x^2 - 9x - 5$

b. $12x^2 - 11x - 5$

c. $10x^2 - 11x - 5$ ✓

d. $10x^2 - 11x - 3$

Combine like terms to get $10x^2 - 11x - 5$.

41 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Subtract: $(2x^2 + 4x + 7) - (6x^2 + x + 3)$

a. $-4x^2 + 3x + 4$ ✓

b. $-2x^2 + 3x + 4$

c. $-4x^2 + 5x + 4$

d. $-4x^2 + 3x + 6$

Combine like terms to get $-4x^2 + 3x + 4$.

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42 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which expression is equivalent to $(a^5)^3$?

a. a^{15} ✓

b. a^8

c. a^9

d. a^{17}

Applying Exponentials, Radicals, Polynomials & Rational Expressions principles, $(a^5)^3 = a^{5 \cdot 3} = a^{15}$

Therefore, the verified correct option is a^{15}.

43 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Add: $(8x^2 - 3x + 5) + (2x^2 - x - 3)$

a. $12x^2 - 4x + 2$

b. $10x^2 - 4x + 4$

c. $10x^2 - 2x + 2$

d. $10x^2 - 4x + 2$ ✓

Combine like terms to get $10x^2 - 4x + 2$.

44 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Add: $(7x^2 - 7x + 4) + (5x^2 + x + 4)$

a. $12x^2 - 6x + 10$

b. $14x^2 - 6x + 8$

c. $12x^2 - 4x + 8$

d. $12x^2 - 6x + 8$ ✓

Combine like terms to get $12x^2 - 6x + 8$.

45 [Challenging] • [MCQ] • Target: Exponential Growth/Decay, Rational Exponents & Polynomials

Which of the following is equivalent to $(x^3)^4$?

a. x^7

b. x^{12} ✓

c. x^{34}

d. x^{81}

Using the power of a power property, $(x^a)^b = x^{ab}$. Therefore $(x^3)^4 = x^{3 \times 4} = x^{12}$.

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